Yield Curve Tension as a Macro Risk Conditioning Variable:

Quantifying Macro Stress For Risk-Aware Equities Trading

Shreyas Shrestha
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Objective:
The U.S. treasury yield curve, which connects interest rates from short-term to long-term government bonds, has historically been representative as a warning sign for economic downturns. When short-term rates exceed long-term rates, the curve "inverts", and every recession in the past 50 years has been preceded by one. I was inspired by this video *"The Chart That Predicts Recessions"*, which piqued my interest in if there was a strong enough relationship to test whether the yield curve could be used as a signal in equity markets. My main research questions were: **can yield curve dynamics serve as a predictive indicator for upcoming equity market volatility, and can such a signal empower a portfolio manager to proactively limit risk exposure prior to significant market downturn?**

Data Collection
For this project, I pulled data across 25 years of daily U.S. Treasury yield rates (3-month through 30-year) from the Federal Reserve's FRED database and SPY daily price data from Yahoo Finance, covering January 2000 through December 2024. After cleaning, and pre-processing the data, we had 6,268 trading days spanning 3 major U.S. recessions, four Federal reserve hiking cycles, and three distinct yield curve inversions. I validated my results on external datasets, pulling from yield curve and equity index data for Germany, the UK, and Canada from the same FRED source, to see if the model could be generalized for any market.

Methodology:
This project employed five analytical layers. Linear correlation testing, Unsupervised Hidden Markov Model regime detection with AIC/BIC model selection, the creation of a four-component Yield Curve Tension Index, Newey-West corrected regression for statistical validation, and a three-state rule-based portfolio overlay backtested over 6,268 trading days with both walk-forward and cross-market validation.

First, I tested out a simple hypothesis: yield curve inversion can directly predict stock market duress. This was quickly invalidated. The correlation between the 10-year minus 2-year spread and forward stock market volatility is 0.071, statistically indistinguishable from 0. Regardless of how I interpreted the results, lagged versions, conditional on inversion, different spread combinations, there was no significant relationship to draw from here.

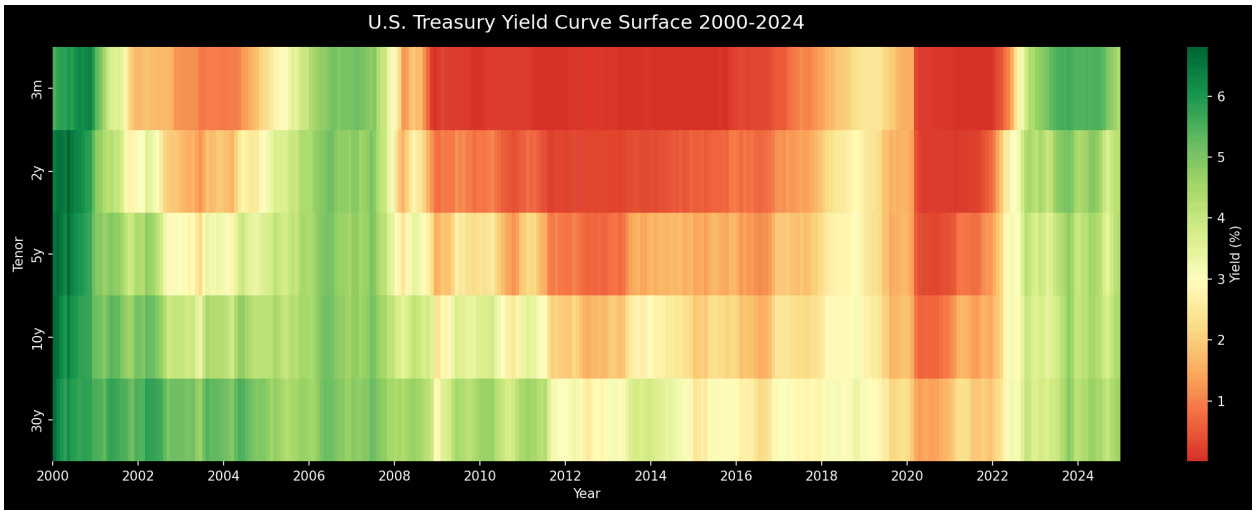
Second, I built an unsupervised machine learning model, a Hidden Markov Model, to learn from the yield curve and volatility data to form natural groupings in the data. No context was provided on recessions, economic history, or any other market disturbances. From this model, we found two main groupings, which I categorized compared to official U.S. recession dates.

- State 1 (Late-Cycle/Pre-Crisis): Flat or inverted curve, characterized by low stock market volatility. During this state, equity markets are still complacent/no-significant changes

- State 2 (Post-Crisis): Steep curve, with elevated volatility. The Fed has often already cut rates but there is still significant financial stress.

My model showed that State 2 captured 94.3% of all recession dates, with simply yield curve and realized volatility data. Both states interestingly enough exhibit near-perfect self-persistence with transition probabilities of 0.999 and 0.998 respectively. This confirms that macro regimes are not just noisy switching processes. I tested both 2-state and 3-state specifications, selecting 2-state on both AIC (42,155 vs 43,283) and BIC, where the 3-state model produced a degenerate third state with zero self-transition probability, confirming two genuine regimes in the data.

Decomposing SPY returns by regime reveals the Late-Cycle state generates a Sharpe ratio of 0.498 versus 0.335 for the Post-Crisis state, a 48% difference on the exact same underlying asset, driven entirely by the lower volatility during the complacent pre-crisis period. However, what was more eye-catching was that the model also showed that it is **not** in fact the moment of the inverted curve which is the dangerous state, but instead when the curve steepens back up and volatility spikes. I considered what could be done to primarily diagnose this, but also provide insights to equities traders to incorporate yield curve tension as a macro conditioning variable.



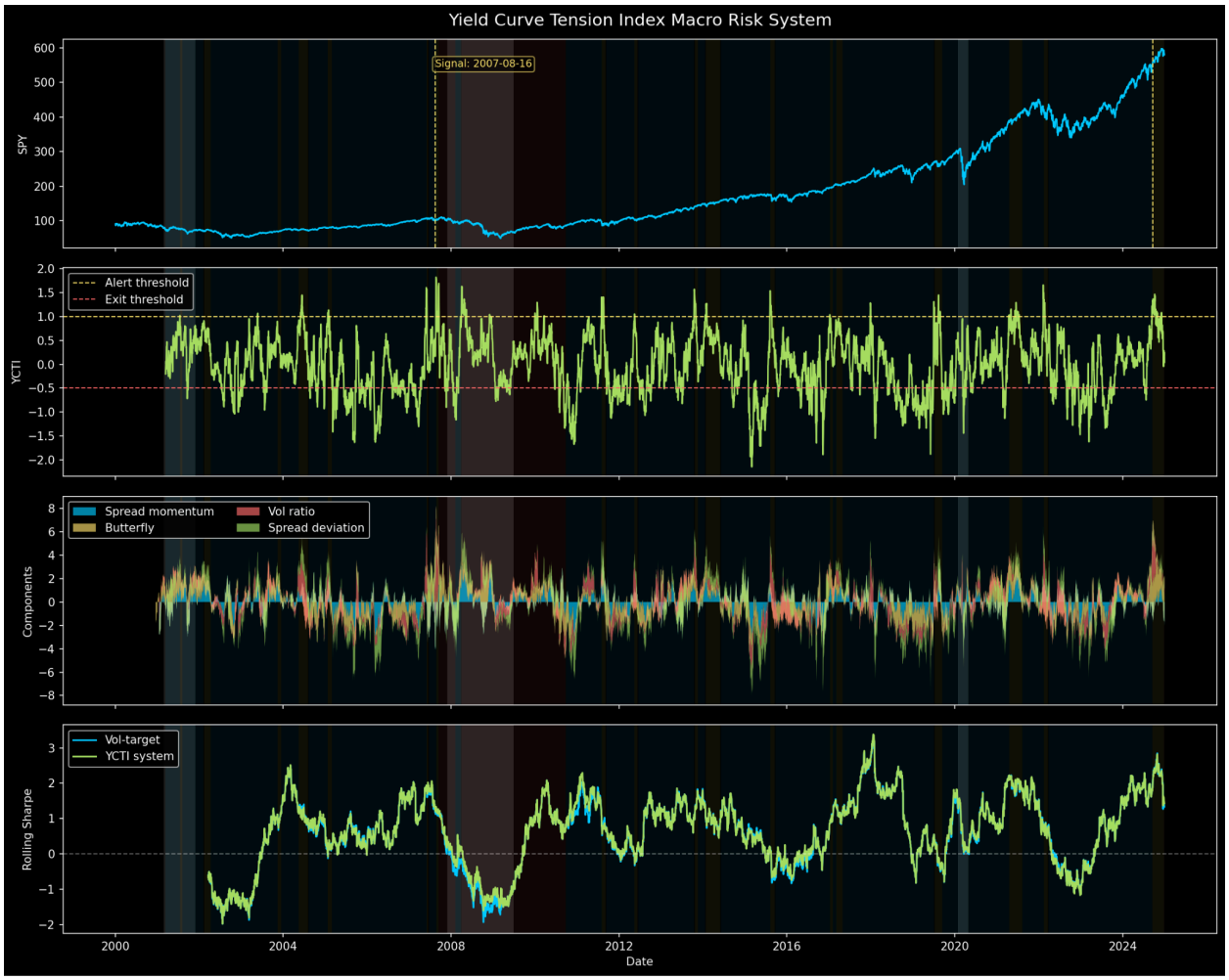
U.S. Treasury yield curve surface 2000-2024

I progressed with the final stage of my project, to build a composite indicator to serve as a framework for risk analysis at the macro level. I formalize this as the *Yield Curve Tension Index*, which combines 4 distinct dimensions of yield curve stress.

- How fast the curve is flattening or steepening
- Mid-curve distortion (the "butterfly spread")
- Whether short-term yield volatility is exceeding long-term yield volatility
- How far the curve has deviated from its recent baseline

I built a rule to then specifically detect the specific pattern which the model highlighted, a prolonged period of curve compression followed by rapid re-steepening, with the tension index still elevated. The idea is that the moment the yield curve “un-inverts” is the moment danger materializes.

This rule fired on **August 16, 2007** exactly 7 days after BNP Paribas froze three investment funds, and **347** days before the bottom hit by the 2008 financial crisis. Now, this is just one data point, but the timing and signal is theoretically grounded: when the Fed starts cutting and the curve re-steepens, it confirms that credit stress has already begun.



YCTI overlay on SPY price 2000-2024.
Signal fired August 16, 2007, 347 days before the GFC trough

I find the Yield Curve Tension Index (YCTI) has a statistically significant relationship with future volatility. To determine this, I ran a formal regression test on whether the YCTI can predict stock market volatility over the next 21 days, applying a statistical correction called Newey-West to get honest standard errors. Why is the Newey-West important? Well, because 21

day forward volatility windows overlap by construction, for example day n shares 20 days with day n+1 which inflates the significance of OLS results.

My results were the following: $\beta = 0.0165$, $t = 2.14$, $p = 0.032$. This is statistically significant at the 5% level. When the YCTI is elevated, next-month realized volatility is meaningfully higher on average. However, it is important to note, I also had an $R^2 = 0.0085$, signifying the macro variable represents less than 1% of forward variability. Thus, it is far better off being used as a tool to increase risk-awareness rather than a direct predictor.

As an additional validation step, I also ran a 252-day rolling OLS estimating whether yield curve spreads predict forward crisis regime probability. There were statistically significant negative betas appearing in 12.6% of estimation windows, mostly around late-cycle transitions, which is consistent with the HMM findings.

To test this index being utilized by an equities trader, I tested a simple three-state risk overlay using the YCTI.

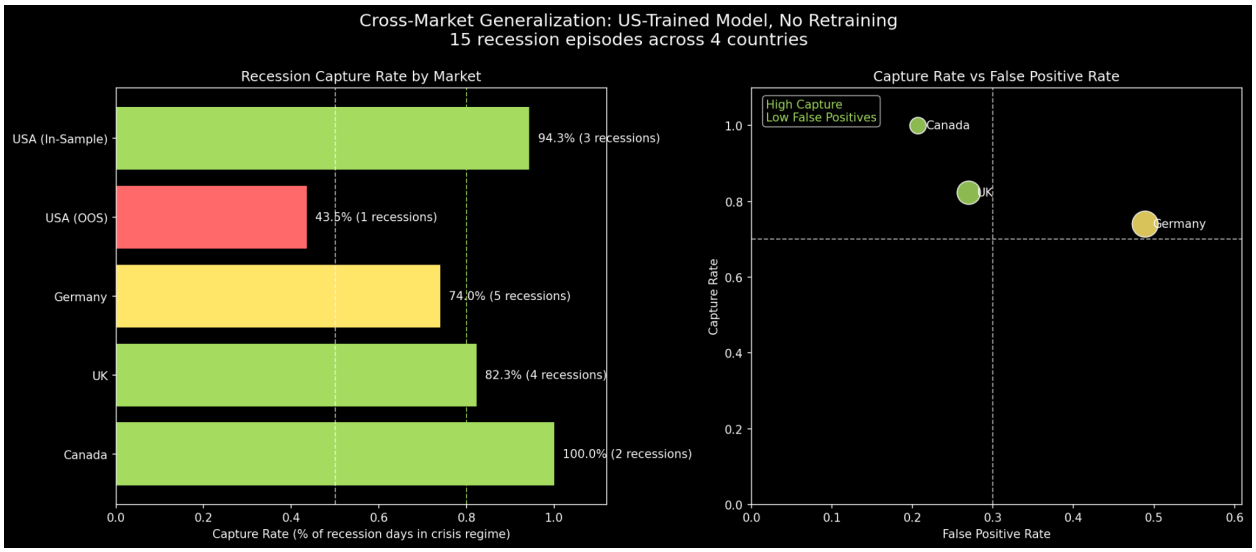
- Normal: Full exposure
- Alert (YCTI > 0.75): 75% exposure
- Defensive (YCTI > 1.50): 50% exposure

Results over the full 2000-2024 sample:

Strategy	Sharpe Ratio	Max Drawdown	Calmar Ratio
Buy & Hold SPY	0.451	-59.6%	0.145
Vol-Targeting	0.596	-27.6%	0.240
YCTI Overlay	0.644	-26.3%	0.253

On a hypothetical $1 million portfolio, the YCTI overlay preserves $92,100 more capital at the worst drawdown point versus vol-targeting alone, and $332,900 more versus buy-and-hold.

Furthermore, I wanted to see how my model would apply across markets to see if results were truly indicative of real-market trends or just picking up signals specific to the U.S. To do so, I tested the U.S.-trained model, without any retraining, to yield curve and equity data from Germany, the UK, and Canada. Across 15 independent recession episodes, I find that this model can be generalized across markets, as shown below.



The cross-market result is our strongest robustness check. A model that generalizes across monetary regimes and financial systems without retraining is capturing something real about how yield curves behave globally, and provides insight into its generalizability. As an additional check for model validity, I trained the HMM exclusively on 2000-2010 data and applied it to 2011-2024 without retraining. Recession capture dropped to 43.5% on the holdout period, which contained only one recession, COVID-19, an exogenous shock with no yield curve precursor. This is a structural limitation of any yield-curve-based model, not a model failure specifically.

Reflection:

Throughout this project, I was able to create a statistically grounded exploratory framework for using yield curve dynamics to condition macro risk assessment. It is a novel composite signal with documented construction, validated regression significance, and cross-market robustness, which I am proud to say. I acknowledge the model and methodology have limitations, as three U.S. recessions is too small a sample to validate timing claims statistically. Furthermore, there is also a realization that exogenous shocks with no yield curve precursor are structurally invisible to this approach. The natural next step is extending validation to UK and German yield curve history back to the 1970s, where 15+ additional recession episodes would provide a more sufficient sample. Also interestingly, the YCTI fired a second signal on September 19, 2024, the date of the first Fed rate cut in the post-COVID hiking cycle, meaning we are in a live and ongoing test of the central hypothesis.

Sources:

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